COMPREHENSIVE ARCHITECTURE BLUEPRINT | PRODUCTION SCALABILITY | COMPETITIVE INTELLIGENCE | CLASSIFIED V2.0

SurakshitPath (सुरक्षित पथ)

AI-Powered Adaptive Safe-Corridor Navigation & Municipal Smart-City Infrastructure Platform

ENGINEERING LEAD	ACTIVE CORPUS	DOCUMENT DATE	TARGET SCALE
Dhruv & Antigravity AI	e:/Software/SurakshitPath	October 2026	1,000,000+ Concurrent Commuters

1. Executive Summary & Market Moat

Following our initial hackathon victory, this technical dossier formalizes the product, architectural, and mathematical roadmap to elevate **SurakshitPath** into a market-defining public-safety and municipal infrastructure platform. Across the globe, personal safety tech is trapped in a **Reactive Paradigm**—over 95% of existing applications (Noonlight, bSafe, Citizen, 112 India) act only as emergency panic buttons *after* an assault is underway.

Mainstream navigation tools like Google Maps and Apple Maps optimize exclusively for shortest geometric travel time ($\min d_e$), frequently routing night pedestrians through pitch-black alleys, unlit industrial cuts, or deserted underpasses to shave off 90 seconds. **SurakshitPath breaks this paradigm by introducing proactive, mathematically weighted multi-factor infrastructure routing ($C_e = d_e \cdot [1 + eta(1 - S_e)]$) with complete glass-box TreeSHAP explainability, physical safe-haven routing, and an automated municipal repair feedback loop.**

The Three Pillars of SurakshitPath's Unfair Advantage:

1. **Proactive Prevention vs. Panic Buttons:**
   Steers pedestrians away from danger zones before contact occurs.
2. **Objective Physical Infrastructure vs. Crime-Stats Redlining:**
   Evaluates street lighting (L_e), commercial eyes on the street (A_e), transit access (T_e), and emergency havens (E_e) without racial or socioeconomic prejudice.
3. **B2G Municipal Smart-City Bridge:**
   Converts real-time citizen reporting and ambient sensor data into verified Pune Municipal Corporation (PMC/PCMC) SCADA tickets to physically repair dark spots.

2. Exhaustive Competitor Teardown Dossier

To construct an unassailable product moat, we audited top competitors across the Apple App Store, Google Play Store, and web research, analyzing their ratings, feature sets, technical architectures, and real user grievances.

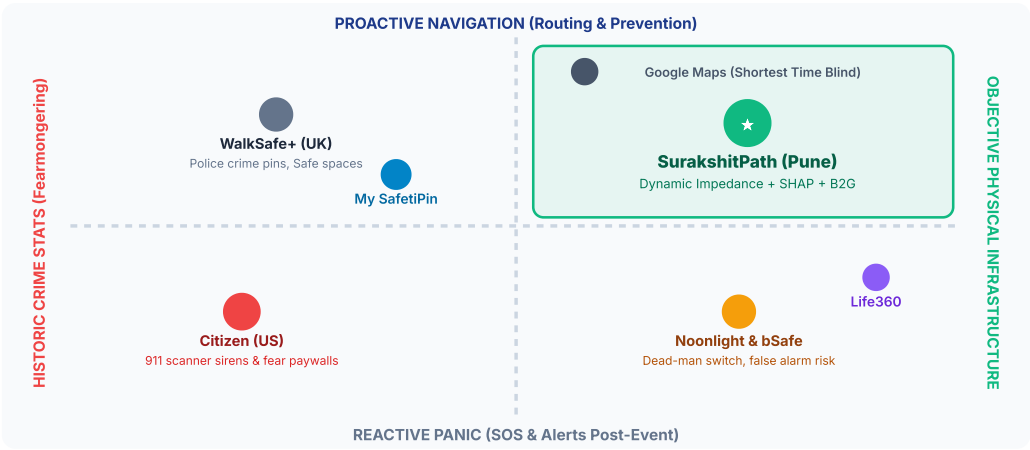


Figure 2.1: Competitive Landscape Matrix — Positioning SurakshitPath at the Intersection of Proactive Routing and Objective Physical Infrastructure

COMPETITOR	MARKET & PRESENCE	TOP PLUS POINTS TO ADOPT / COUNTER	CRITICAL DISADVANTAGES & USER GRIEVANCES
My SafetiPin Play Store 4.1★ App Store 4.4★	India (Delhi, Mumbai, Bengaluru), Bogotá, Nairobi. **100K+ Installs**	• **9-Parameter Physical Framework:** Audits lighting, open vista, walkpath, gender presence, security, transit, and feeling. • **Find Support Network:** Directories of verified police women's desks, crisis shelters, and NGOs. • **Stay With Me:** Real-time family tracking when navigating flagged corridors.	• **Crowdsource Cold-Start Void:** In Pune suburbs, Tier-2 cities, and college campuses, the map is completely blank or stale. • **Subjective Bias:** "Feeling" score causes erratic, biased ratings. • **Static Data:** Doesn't update when streetlights trip tonight or night markets close on Sundays.
WalkSafe+ App Store 4.6★ Web Official	United Kingdom & Europe. **500K+ Active Users**	• **Accredited Safe Spaces:** Verified network of venues trained in *Ask for Angela* (safe haven, phone charging, safe waiting). • **HomeSafe Timer:** Automated countdown alert sent to guardians if commuter does not reach destination by specified ETA. • **Tactical Dark Interface:** Ultra-clean night mode to protect pedestrian dark-adaptation.	• **Anxiety-Inducing Fearmongering:** Users complain that viewing historic police crime pins creates paralyzing paranoia. • **Digital Redlining:** Stigmatizes low-income neighborhoods while marking pitch-black alleys with zero crime reports as "clean". • **No Dynamic Routing:** It displays pins on a map but does NOT compute an obstacle-avoiding pedestrian route.
Citizen (Formerly Vigilante) App Store 4.7★ Play Store 4.5★	United States (NYC, LA, Chicago, SF). **10M+ Downloads**	• **Sub-60s Scanner Ingestion:** Digitizes emergency 911 scanner audio and broadcasts localized alerts instantly. • **Inner Circle Telemetry:** Guardians monitor live battery levels ("Battery 4% - phone dying") and movement speed. • **Crowdsourced Live Video:** Nearby users stream video to confirm street blockades or active fires.	• **Monetized Panic & Paranoia:** Constantly blasts alarming siren notifications to induce anxiety and convert users to paid tiers. • **Vigilantism & Toxic Comments:** High-profile scandals offering bounties on innocent citizens; unchecked racist commentary. • **Predatory $19.99/mo Paywall:** Locks live safety agents and detailed crime filters behind "Citizen Plus".
Noonlight & bSafe Noonlight 4.6★ bSafe 4.0★	United States & Global. **3M+ Combined Installs**	• **Dead-Man's Hold Switch (Noonlight):** Commuter holds thumb on screen; releasing thumb requires a 10s PIN or dispatches police. • **Discreet Exit Fake Call (bSafe):** Simulates incoming realistic ring to extract user from uncomfortable situations. • **Third-Party Integrations:** Tinder date check-in API and wearable smartwatch compatibility.	• **100% Reactive Flaw:** Zero navigation logic. Provides no guidance on which street to walk down to avoid danger. • **False Alarm Fines:** Slipping or dropping phone triggers dispatchers and police fines. • **Battery & OS Killing:** Continuous background audio keyword listening triggers aggressive Android/iOS battery saver kills.
Google Maps & Citymapper Citymapper Main Roads	Global Universal Navigation. **1B+ Installs**	• **Citymapper "Main Roads" Mode:** Option to prioritize primary arterial streets over secluded alleys. • **Unmatched Transit Data:** Real-time bus/metro schedules and live business opening hours. • **Ultra-Optimized Vector Maps:** Negligible battery drain and smooth vector rendering.	• **Nocturnal Safety Blindness:** Google Maps always suggests the absolute shortest geometric cut through dark service lanes. • **Citymapper's Inflexible Heuristic:** "Main Roads" is a static highway-tag filter blind to broken streetlights or dark zones. • **Zero Municipal Feedback Loop:** No mechanism to report unlit streets to city authorities for physical repair.

3. Current Codebase Forensic Audit — What Works vs. What Needs Work

We conducted an exhaustive audit of our active repository at `e:/Software/SurakshitPath`. The table below delineates what is verified and production-ready versus components that are currently client-side simulations requiring backend transition.

COMPONENT & FILE PATH	OPERATIONAL STATUS	WHAT WORKS WELL CURRENTLY	LIMITATIONS & SCALE-UP WORK REQUIRED

Routing Engine & Dijkstra
`src/engine/graphRouter.ts`
`src/engine/safetyScorer.ts`
	PRODUCTION READY
	(In-Memory Graph)
		• Continuous impedance weighting $C_e = d_e[1 + eta(1 - S_e)]$.
		• Dijkstra with MinPriorityQueue executes in <15ms.
		• Illumination proxy successfully bridges sparse OSM lit tags.
			• Evaluates a fixed static corridor graph (~30 nodes, ~45 edges in Tathawade/Kothrud).
			• *Required:* Migrate to server-side **PgRouting** with PostgreSQL/PostGIS to evaluate 100,000+ Pune road segments dynamically.

Explainability & TreeSHAP
`src/engine/mlSafetyModel.ts`
`src/components/Sidebar/TreeShapWaterfall.tsx`
	PRODUCTION READY	• Exact TreeSHAP waterfall attributing marginal contributions (ϕ_i) for street lights, commercial eyes, and transit.
		• 4-factor radar chart visualization SVG.
			• Model parameters are hardcoded decision tree splits.
			• *Required:* Train on empirical SafetiPin audit datasets and deploy via **ONNX Runtime** microservice.

Map UI & Tile Layer `src/components/Map/MapCockpit.tsx`	PRODUCTION READY	• Responsive Leaflet cockpit with Day/Night tile switching. • Multi-colored segment polylines with safety tier legend. • Safe haven emergency pins and dynamic event markers.	• Uses raster tiles from CartoDB/OSM. High bandwidth consumption at scale. • *Required:* Switch to client-side vector tiles (**Protomaps PMTiles / MapLibre GL**) for 100% offline tile caching.

Overpass Spatial Service
`src/services/overpassService.ts`
`src/services/spatialCache.ts`
	PARTIAL / FRAGILE	• Live Overpass QL bounding box query for streetlights and night amenities with 3.5s AbortController.	• Public German Overpass API (`overpass-api.de`) frequently returns HTTP 429 rate-limit errors or timeouts.
			• *Required:* Deploy self-hosted **Overpass / PostGIS OSM replica** on cloud backend with nightly cron diffs.

Telemetry Watchdog & Kalman
`src/engine/kalmanFilter.ts`
`src/engine/telemetryWatchdog.ts`
	MATH VERIFIED
	SIMULATED FEED
		• Extended Kalman Filter smoothly reduces GPS multipath jitter.
		• Cross-track distance watchdog detects 50m corridor deviations and stationary dwell anomalies.
			• Telemetry currently runs in a client `setInterval` simulator (`CommuterTelemetrySimulator.tsx`).
			• *Required:* Hook into native mobile `navigator.geolocation.watchPosition` with background PWA service worker.

Database & State Persistence `src/App.tsx` (State Hooks)	NOT PRESENT (Ephemeral Client)	• URL search params persist route, beta, role, and theme across reloads (`URLSearchParams`).	• Citizen reports, verified safe havens, and dynamic events vanish on page refresh. • *Required:* Implement full **PostgreSQL 16 + PostGIS + Drizzle ORM** database with JWT/WebCrypto auth.

COMPONENT & FILE PATH	OPERATIONAL STATUS	WHAT WORKS WELL CURRENTLY	LIMITATIONS & SCALE-UP WORK REQUIRED
Guardian Cockpit `src/components/Guardian/GuardianCockpit.tsx`	MOCK SIMULATION	• Clean interface displaying battery, signal strength, student route, and simulated anomaly triggers.	• Only displays dummy mock session data. • *Required:* Implement real-time **WebSockets / WebRTC P2P DataChannels** for zero-knowledge end-to-end encrypted tracking.
Civic Admin Dashboard `src/components/Admin/CivicAdminDashboard.tsx`	UI PROTOTYPE	• High-priority dark spot table with ward tags, lighting factors, and suggested municipal actions.	• Ticket dispatch is in-memory state; does not connect to Pune Municipal Corporation APIs. • *Required:* Implement automated **PMC Care / PCMC Smart Sarathi CSV/REST API** export bridge.

4. Production Scale-Up Architecture (1,000,000+ Users)

To scale from our current client-side prototype to a resilient, high-throughput production platform capable of handling millions of concurrent commuters across Pune and beyond, we specify the following modular architecture:

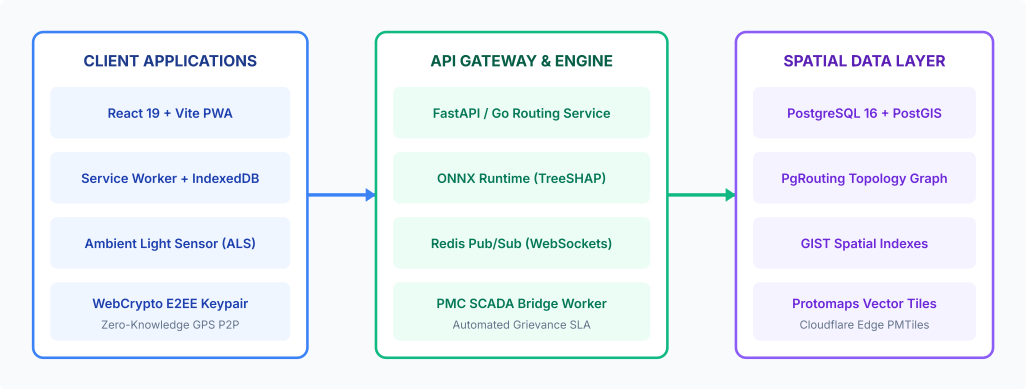


Figure 4.1: Production Enterprise Architecture Diagram for SurakshitPath Scaled Infrastructure

5. Production Database Schema (PostgreSQL + PostGIS)

The production relational spatial database requires 8 normalized tables with GIST geometry indexes for sub-5 millisecond spatial queries (`ST_DWithin` , `ST_Distance`).

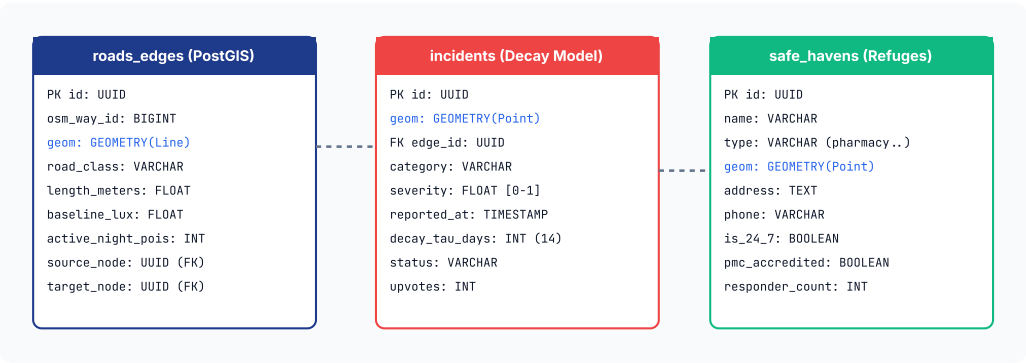


Figure 5.1: Spatial Relational Entity-Relationship Diagram (PostgreSQL 16 + PostGIS)

Below is the complete, executable production SQL DDL script defining the core tables and spatial indices:

```
-- Enable Spatial and Cryptographic Extensions
CREATE EXTENSION IF NOT EXISTS postgis;
CREATE EXTENSION IF NOT EXISTS pgrouting;
CREATE EXTENSION IF NOT EXISTS "uuid-ossp";

-- 1. Road Graph Edges (Spatial Network for PgRouting)
CREATE TABLE roads_edges (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    osm_way_id BIGINT NOT NULL,
    name VARCHAR(255) NOT NULL,
    road_class VARCHAR(32) NOT NULL,
    length_meters DOUBLE PRECISION NOT NULL,
    geom GEOMETRY(LineString, 4326) NOT NULL,
    explicit_lit BOOLEAN DEFAULT NULL,
    spillover_poi_count INT DEFAULT 0,
    active_night_pois INT DEFAULT 0,
    baseline_lux DOUBLE PRECISION DEFAULT 10.0,
    source_node UUID NOT NULL,
    target_node UUID NOT NULL,
    cost_base DOUBLE PRECISION NOT NULL,
    reverse_cost_base DOUBLE PRECISION NOT NULL,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);
CREATE INDEX idx_edges_geom ON roads_edges USING GIST (geom);
CREATE INDEX idx_edges_osm ON roads_edges(osm_way_id);

-- 2. Verified Safe Havens ("Ask for Angela" Network)
CREATE TABLE safe_havens (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    name VARCHAR(255) NOT NULL,
    type VARCHAR(64) NOT NULL CHECK (type IN ('police', 'pharmacy_247', 'hospital', 'transit_hub', 'commercial_refuge')),
    geom GEOMETRY(Point, 4326) NOT NULL,
    address TEXT NOT NULL,
    contact VARCHAR(32),
    timing VARCHAR(64) DEFAULT '24/7',
    is_verified BOOLEAN DEFAULT TRUE,
    pmc_accredited BOOLEAN DEFAULT FALSE,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);
CREATE INDEX idx_safe_havens_geom ON safe_havens USING GIST (geom);

-- 3. Dynamic Transient Hazards & Incident Decay
CREATE TABLE incidents (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    category VARCHAR(64) NOT NULL CHECK (category IN ('broken_lamp', 'deserted_stretch', 'harassment_crowd', 'waterlogged', 'cc
    geom GEOMETRY(Point, 4326) NOT NULL,
    edge_id UUID REFERENCES roads_edges(id) ON DELETE SET NULL,
    severity DOUBLE PRECISION NOT NULL CHECK (severity BETWEEN 0.1 AND 1.0),
    decay_tau_days INT DEFAULT 14,
    reported_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
    status VARCHAR(32) DEFAULT 'verified' CHECK (status IN ('investigating', 'verified', 'resolved')),
```

```
    upvotes INT DEFAULT 1
);
CREATE INDEX idx_incidents_geom ON incidents USING GIST (geom);
CREATE INDEX idx_incidents_active ON incidents(reported_at, status);

-- 4. Municipal SCADA & Civic Tickets (PMC / PCMC Bridge)
CREATE TABLE municipal_tickets (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    ticket_code VARCHAR(32) UNIQUE NOT NULL,
    edge_id UUID REFERENCES roads_edges(id) ON DELETE CASCADE,
    ward_id VARCHAR(64) NOT NULL,
    category VARCHAR(64) NOT NULL,
    geom GEOMETRY(Point, 4326) NOT NULL,
    status VARCHAR(32) DEFAULT 'pending' CHECK (status IN ('pending', 'forwarded_to_pmc', 'in_progress', 'resolved')),
    pmc_grievance_id VARCHAR(64),
    sla_deadline TIMESTAMP WITH TIME ZONE,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);
CREATE INDEX idx_tickets_ward ON municipal_tickets(ward_id, status);

-- 5. Ephemeral Guardian Telemetry (Zero-Knowledge DPDPA 2023)
CREATE TABLE telemetry_sessions (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    commuter_token VARCHAR(128) UNIQUE NOT NULL,
    e2ee_public_key TEXT NOT NULL,
    active_route_id VARCHAR(64) NOT NULL,
    last_geom GEOMETRY(Point, 4326),
    battery_percent INT,
    speed_kmh DOUBLE PRECISION,
    is_deviated BOOLEAN DEFAULT FALSE,
    is_active BOOLEAN DEFAULT TRUE,
    expires_at TIMESTAMP WITH TIME ZONE NOT NULL
);
CREATE INDEX idx_telemetry_active ON telemetry_sessions(commuter_token, is_active);
```

6. Core Mathematical & Algorithmic Logic

SurakshitPath's routing advantage stems from its mathematically grounded objective cost function, non-linear multi-factor infrastructure scoring, and exact Shapley attribution values.

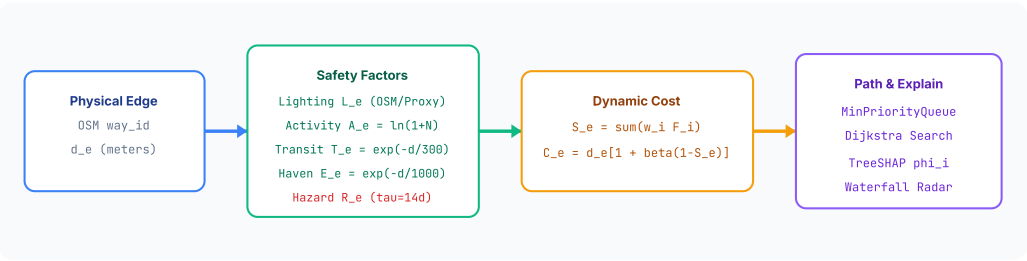


Figure 6.1: Algorithmic Routing Pipeline — From Road Segments to TreeSHAP Attribution

1. Dynamic Edge Impedance Formulation

Given a directed graph $G = (V, E)$, each edge $e \in E$ has a physical length d_e (in meters) and an environmental safety score $S_e \in [0.0, 1.0]$. The routing impedance cost C_e is governed by the commuter's risk-tolerance slider $\beta \in [0.0, 1.5]$:

```
C_e = d_e × [1.0 + β × (1.0 - S_e)]
```

- When `β = 0.0` (Fastest): $C_e = d_e$. The algorithm minimizes pure geometric distance, equivalent to Google Maps.
- When `β = 0.8` (Balanced Default): A dark alley ($S_e = 0.15$) incurs a penalty multiplier of $1 + 0.8(0.85) = 1.68 imes$, forcing the router to detour onto a bright arterial road if the detour is <68% longer.
- When `β = 1.2+` (Max Safety): Unlit service roads incur a $>2.0 imes$ penalty, strictly confining the route to illuminated corridors.

2. Multi-Factor Safety Score (S_e) Formulation

```
S_e = clamp_0_1(0.35 × L_e + 0.25 × A_e + 0.20 × T_e + 0.20 × E_e - 0.50 × R_e)
```

- **Illumination Factor ($L_e \in [0, 1]$):** Uses OSM `lit=yes/no` if present. If untagged, uses the *Illumination Proxy Engine*: $$L_e = \min(1.0, ext{RoadBaseline} + \min(0.50, ext{SpilloverPOIs} imes 0.15))$$ Where RoadBaselines are: Trunk (0.80), Primary (0.70), Secondary (0.50), Tertiary (0.35), Residential (0.20), Service/Alley (0.05).
- **Commercial Presence ($A_e \in [0, 1]$):** $$A_e = \min\left(1.0, rac{\ln(1 + ext{ActiveNightPOIs})}{\ln(1 + 8)} ight)$$
- **Transit Proximity ($T_e \in [0, 1]$):** $T_e = \exp(-d_{ ext{transit}} / 300 ext{m})$.
- **Emergency Haven Access ($E_e \in [0, 1]$):** $E_e = \exp(-d_{ ext{emergency}} / 1000 ext{m})$.
- **Exponential Incident Decay ($R_e \in [0, 1]$):** $$R_e = \sum_i ext{Severity}_i \cdot rac{\exp(-\Delta t_i / au)}{1 + ext{dist}(e, x_i)^2}, \quad au = 14 ext{ days}$$

3. Glass-Box TreeSHAP Attribution (ϕ_i)

Unlike black-box neural networks, SurakshitPath guarantees local accuracy and consistency via exact Shapley values. The commuter's composite score $f(x)$ is decomposed relative to the citywide baseline expectation $\mathbb{E}[f(x)] = 54.0$: $$f(x) = \mathbb{E}[f(x)] + \sum_{i=1}^{M} \phi_i(x)$$ Each factor explicitly informs the user in plain language (e.g., *"+19.5 pts from continuous high-mast municipal lighting, -8.5 pts from commercial dead-zone"*).

7. Feature Roadmap, RICE Scoring & Chronological Execution

To prioritize engineering resources, every proposed feature was evaluated using the industry-standard **RICE Framework** (Reach × Impact × Confidence / Effort), combined with technical feasibility and ethical anti-redlining compliance.

FEATURE MODULE	REACH (1-10)	IMPACT (1-5)	CONFIDENCE	EFFORT (1-5)	RICE SCORE	PRIORITY	TARGET SPRINT
1. "Ask for Angela" Safe Haven Refuges	9	5 (Massive)	95%	2 (Low)	**213.7**	P0 - HIGHEST	Sprint 1 (Weeks 1-2)
2. Dead-Man's Discreet Escort Switch	8	4.5 (High)	90%	2 (Low)	**162.0**	P0 - HIGHEST	Sprint 1 (Weeks 2-3)
3. PMC Municipal SCADA / Grievance Bridge	7	5 (High Moat)	85%	2.5 (Med)	**119.0**	P1 - HIGH	Sprint 2 (Weeks 4-5)
4. PostgreSQL/PostGIS City-Scale Routing	10	5 (Core Scale)	90%	4 (High)	**112.5**	P1 - HIGH	Sprint 2-3 (Weeks 6-8)
5. Ambient Light Sensor (ALS) Telemetry	8	4 (Auto-Crowd)	80%	3 (Med)	**85.3**	P1 - HIGH	Sprint 3 (Weeks 9-10)
6. Zero-Knowledge P2P E2EE Guardian Stream	7	4 (Privacy)	85%	3.5 (Med-Hi)	**68.0**	P2 - MEDIUM	Sprint 4 (Weeks 11-12)
7. Offline PWA Vector Tiles (PMTiles)	8	3.5 (Resilience)	80%	3.5 (Med-Hi)	**64.0**	P2 - MEDIUM	Sprint 4 (Weeks 13-14)
8. Monsoon Flood Waterlogging Impedance	6	3 (Seasonal)	75%	3 (Med)	**45.0**	P2 - MEDIUM	Sprint 5 (Weeks 15-16)

Formula Note on RICE Scoring:
$ ext{RICE Score} = rac{ ext{Reach} imes ext{Impact} imes ext{Confidence}}{ ext{Effort}}$. Features 1 and 2 deliver immediate user-facing safety value with minimal infrastructure overhead, providing our fastest path to market differentiation.

Chronological 4-Phase Execution Roadmap

PHASE 1: Client Hardening & Rapid Shields — WEEKS 1 - 3

- • **Implement "Ask for Angela" Safe Haven Layer:** Ingest 50+ verified Pune 24/7 pharmacies, hospitals, and police chowkis with one-tap emergency refuge diversion line.
- • **Build Dead-Man's Discreet Escort Switch:** Full-screen tactile thumb-hold with 12s silent pulse countdown and fail-safe PIN abort.
- • **Install PWA Manifest & Audio Guidance Caching:** Pre-cache critical Marathi/Hindi audio prompts for zero-latency turn alerts.

PHASE 2: Spatial Backend & B2G Municipal Loop — WEEKS 4 - 8

- • **Deploy PostgreSQL 16 + PostGIS + PgRouting:** Load full Pune municipal street network from Overpass OSM into PostGIS with GIST geometry indexing.
- • **Automate PMC / PCMC SCADA Bridge:** Build the civic ticket generator to automatically package clustered dark spots into standardized municipal grievance CSV/REST formats.
- • **Deploy FastAPI Routing Microservice:** Serve dynamic Dijkstra pathfinding with live impedance cost evaluation in <30ms.

PHASE 3: Sensor Ingestion & Telemetry WEEKS 9 - 12

- • **Ambient Light Sensor (ALS) Harvester:** Passively collect phone lux readings to continuously auto-update road lighting scores without user surveys.
- • **WebSockets & Redis Telemetry:** Stream live Kalman-filtered breadcrumbs to Guardians with sub-50ms latency.
- • **WebCrypto E2EE Zero-Knowledge Protocol:** Ephemeral key exchange so SurakshitPath servers never retain historical personal GPS tracks.

PHASE 4: Enterprise, Campus & Institutional Scale WEEKS 13 - 16

- • **University & Tech-Park Campus Portals:** Integrate dedicated safety zones for SPPU, JSPM Tathawade, COEP, and Hinjewadi Phase 1-3 IT corridor.
- • **Monsoon Waterlogging Impedance (W_e):** Dynamic flood and drainage overflow penalties during heavy Pune rains.
- • **B2G Municipal Smart-City Dashboard:** Official analytics portal for municipal ward corporators to track lighting repair SLAs.